WORKED SOLUTION

STEP I 1996, Question 11

Mechanics • Projectile Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

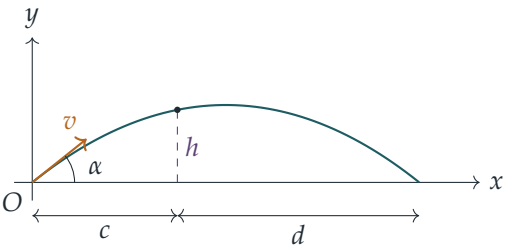
step.maths.org/questions/96-s1-q11

The key idea

The path of a projectile is a parabola, and we know both places where it meets the ground: $x = 0$ and $x = c + d$. A quadratic is pinned down by its roots up to one constant, so the trajectory must be $y = kx(c + d - x)$ for some k. The point (c, h) fixes k, and comparing with the standard trajectory equation turns k into $\tan\alpha$ and then into v^2.

Solution

We model the particle as a point mass moving freely under gravity alone (no air resistance), and take axes with O at the origin, x horizontal in the direction of motion and y vertically upwards. Write $R = c + d$ for the range and $T = \tan\alpha$.



The angle of projection

With initial velocity components $v\cos\alpha$ and $v\sin\alpha$, the position after time t is $x = vt\cos\alpha$, $y = vt\sin\alpha - \frac{1}{2}gt^2$. Eliminating $t = x/(v\cos\alpha)$ gives the trajectory

$$y = x\tan\alpha - \frac{gx^2}{2v^2\cos^2\alpha}.$$

The particle lands at $x = R$, so $y = 0$ there. Since $R \neq 0$ we may divide by R:

$$\frac{g}{2v^2 \cos^2 \alpha} = \frac{\tan \alpha}{R}. \qquad (*)$$

Substituting this back into the trajectory,

$$y = Tx - \frac{Tx^2}{R} = \frac{T\,x(R-x)}{R}.$$

Ask yourself Why is it worth replacing $g/(2v^2 \cos^2 \alpha)$ before using the point (c, h)?

Because $(*)$ removes v completely, leaving an equation in the single unknown T. Geometrically, we have written the parabola in factorised form with its roots 0 and R showing. Now the point (c, h) lies on the path, and $R - c = d$, so

$$h = \frac{T\,cd}{c+d}, \qquad \text{that is,} \qquad \tan \alpha = \frac{h(c+d)}{cd},$$

as required.

The speed of projection

Rearranging $(*)$ and using $\sec^2 \alpha = 1 + \tan^2 \alpha$,

$$v^2 = \frac{gR}{2T \cos^2 \alpha} = \frac{gR(1+T^2)}{2T} = \frac{g}{2}\left(\frac{R}{T} + RT\right).$$

From the first result, $\dfrac{R}{T} = \dfrac{cd}{h}$ and $RT = \dfrac{(c+d)^2 h}{cd}$. Hence

$$v^2 = \frac{g}{2}\left(\frac{cd}{h} + \frac{(c+d)^2 h}{cd}\right),$$

which completes the question.

Checks

- By the AM–GM inequality, $\frac{R}{T} + RT \geq 2R$, so $v^2 \geq gR$, with equality exactly when $T = 1$. This is the familiar fact that the greatest range for a given speed is v^2/g, achieved at 45°.
- If the point (c, h) is the top of the flight, then $c = d$ and $\tan \alpha = 2h/c$, which matches the standard result that the maximum height is a quarter of the range times $\tan \alpha$.
- Numerically, with $g = 9.81$, $c = 3$, $d = 5$ and $h = 2$ the formulae give $\tan \alpha = \frac{16}{15}$ and $v^2 \approx 78.64$. Stepping the motion forward in small time steps, the particle passes through height 2.000 at $x = 3$ and lands at $x = 8.000$.

Common slips

- Forgetting the $\cos^2\alpha$ in the trajectory equation, or replacing $\sec^2\alpha$ by $1+\tan\alpha$ instead of $1+\tan^2\alpha$.
- Trying to solve the two equations "height h at $x=c$" and "height 0 at $x=c+d$" for v and α directly. This works, but the algebra is much heavier than using $(*)$ to eliminate v first.

Beyond STEP

The trick of writing a quadratic in terms of its roots is a small case of a big idea: a polynomial is determined by its zeros up to a constant, which leads on to interpolation and to the factor theorem over the complex numbers. In mechanics, letting α vary for fixed v gives a family of parabolas whose envelope is the "parabola of safety", the boundary of every point the projectile can reach. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectile motion and its extensions (including air resistance) at first-year university level.

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